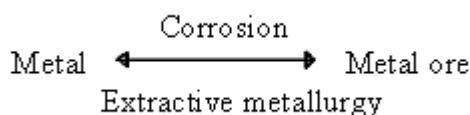


MODULE-3

Corrosion Science and E-waste Management *Corrosion Science*

CORROSION:

Defⁿ: Corrosion is a destruction reaction taking place spontaneously by chemical or electrochemical reaction of the metal by the surrounding environment on the surface of the metal.



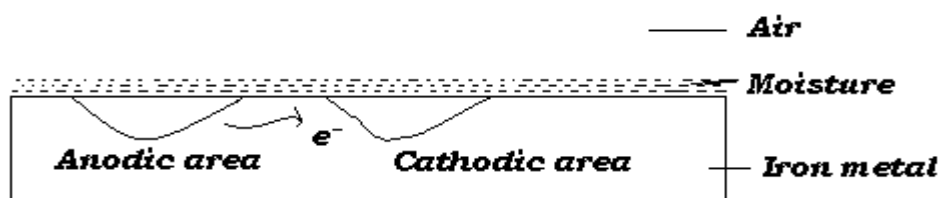
A slow and continuous process of destruction of metallic substance through spontaneous chemical and electrochemical attack by its environment starting at its surface is called corrosion.

Reason: Pure isolated metals will have higher energy content than their corresponding ores and their compounds hence they have a natural tendency to revert back to combined form by combining with the contents of the environment such as gases, liquids, moisture etc. when exposed. This is the basic reason for corrosion.

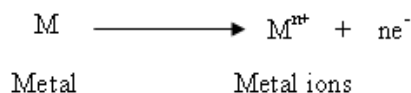
Electrochemical theory of corrosion

According to electrochemical theory, corrosion of metals occurs due to the following changes, when the metals are exposed to the environment.

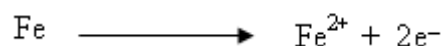
- 1) A large number of minute galvanic cells are formed which acts as anodic and cathodic areas.
- 2) At anode the metal undergoes oxidation and electrons are liberated which migrates towards cathodic region
- 3) Oxygen of the atmosphere undergoes reduction at cathodic area in the presence of moisture forming hydroxyl ions at the cathode



Anodic reactions: At anode the metal undergoes oxidation-liberating electrons



Ex: when iron is exposed to the environment it undergoes oxidation as



Cathodic reactions: The electrons released at anode migrate to the cathodic area and reduces oxygen to hydroxyl ions.

The different cathodic reactions are,

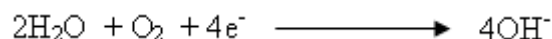
a) In acidic medium: In acidic medium and in the absence of oxygen, hydrogen ions are reduced to hydrogen gas



b) In alkaline and in the absence of O_2 . If the solution is alkaline and in the absence of oxygen the cathodic reaction is,



c) In neutral and aerated medium: when the solution is neutral and aerated, hydroxyl ions are formed as follows.



d) Formation of corrosion product: The hydroxyl ions migrate towards anode and react with metal ions (M^{n+} ions) and forms corrosion product. In the case of iron OH^{-} reacts with Fe^{2+} ions and forms an insoluble hydrated ferric oxide known as brown rust.



If i_{corr} (corrosion current) is the current flowing,

Then Rate of corrosion = i_{corr}/nF grams per second. n =charge on the ions, F =Faraday constant

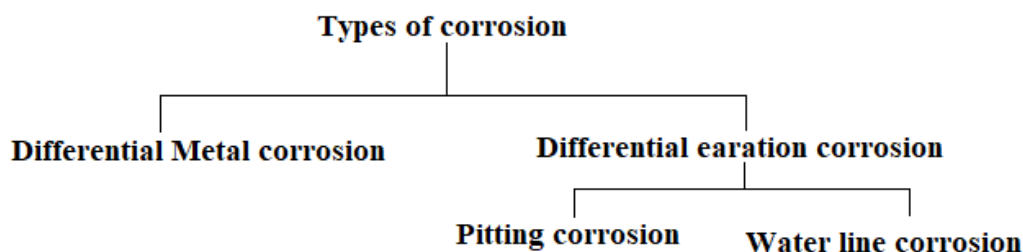
The total current due to the cathodic reaction ($\sum i_c$) must be equal but opposite in sign, to the total current flowing out due to the anodic reaction ($-\sum i_a$).

$$i_{\text{corr}} = -\sum i_a = \sum i_c$$

Galvanic Series: The arrangement of elements in the order of their standard reduction potential is referred to as emf or electrochemical series. Such an arrangement of few elements given in the table.

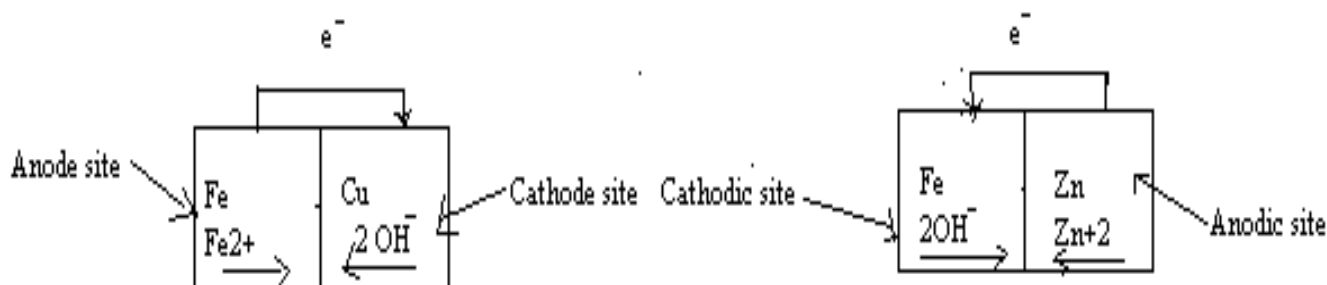
Mn^{+}/Mn	$E^{\circ}(\text{V})$	Mn^{+}/Mn	$E^{\circ}(\text{V})$
Li^{+}/Li	-0.305	Sn^{2+}/Sn	-0.14
K^{+}/K	-2.93	Pb^{2+}/Pb	-0.126
Ca^{+}/Ca	-2.87	H^{+}/H_2	0.000
Na^{+}/Na	-2.71	Cu^{2+}/Cu	0.34
Zn^{2+}/Zn	-0.76	Ag^{+}/Ag	0.80

Fe^{2+}/Fe	-0.44	Hg^{2+}/Hg	0.85
Cd^{2+}/Cd	-0.40	Pt^{2+}/Pt	1.20
Ni^{2+}/Ni	-0.236	Au^{3+}/Au	1.38



1). Differential metal corrosion:

When two dissimilar metals are in direct contact with one another and exposed to a corrosive conducting medium, the metal higher up in the electrochemical series behaves as anode and suffers from corrosion, where as the metal lower in the electrochemical series become cathode with respect to the other metal and is protected from corrosion. This type of corrosion is also known as Galvanic corrosion.



When Zn & Cu metals are electrically connected and exposed to an electrolyte, Zn metal higher in galvanic series forms anode and suffers from corrosion whereas Cu lower in electrochemical series forms cathode hence protected from corrosion. The extent of corrosion depends on the difference in the electrode potential of the two metals and also their relative areas exposed, if the potential difference between the electrodes is high, greater the extent of corrosion.

Eg: 1). Brass taps fixed to an iron pipe. 2). Steel screws in a brass marine hard ware.

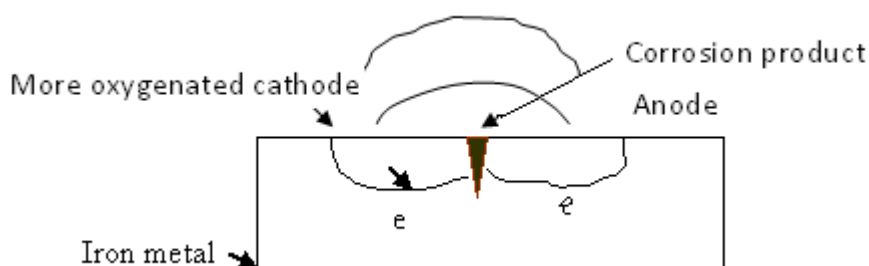
2). Differential Aeration Corrosion:

When a metal surface is exposed to differential air or oxygen concentrations forms differential aeration cell. The more oxygenated part of the metal behaves as cathode and the less oxygenated part become anode. Differential aeration of metal causes a flow of current called the differential current & the corrosion is called as differential aeration corrosion.

This phenomenon can be explained by two types of differential aeration corrosion. They are.

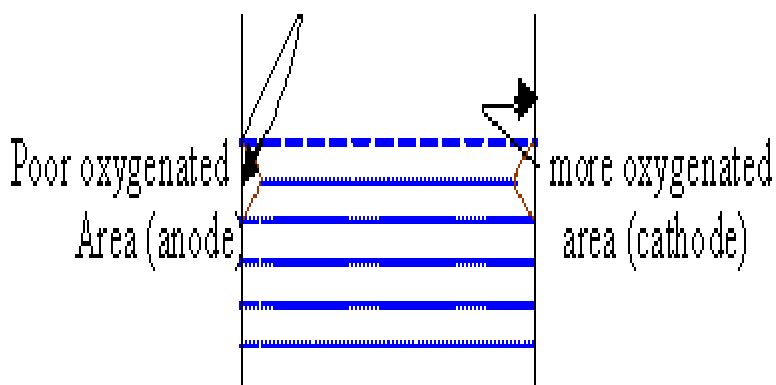
a. Pitting Corrosion

b. Water line corrosion

a. Pitting Corrosion:• **Pitting**

corrosion is a localised accelerated attack in which only small areas of the metal surface are attacked whilst the remainder is largely unaffected. This localised attack results in pitting. The pits may initiate and propagate to a certain depth resulting in the formation of cavities and becomes inactive.

- Pitting is very destructive and frequently ruins the tubes, pipes etc.
- Pitting is due to breakdown or cracking of the protective film on a metal at specific points.
- The presence of impurities like sand, dust, scale, etc., on the surface of metal leads to pitting.
- Pitting corrosion is due to the formation of differential aeration cell.
- This attack becomes more intensified with time.

b. Water line corrosion:

- Differential aeration accounts for the corrosion of metals partially immersed in a solution, just below the water line. This type of differential aeration corrosion is also known as **water line corrosion**.
- Consider a steel tank containing water. The maximum corrosion takes place along a line just beneath the level of water meniscus. The area above the waterline is highly

oxygenated and acts as the cathodic and completely unaffected by corrosion. (Eg. Marine plants attacking themselves in the sides).

Corrosion control

Corrosion can be controlled by preventing the formation of galvanic cells the common methods used to control corrosion of metals are as follows.

1. Metal coatings:-Galvanization

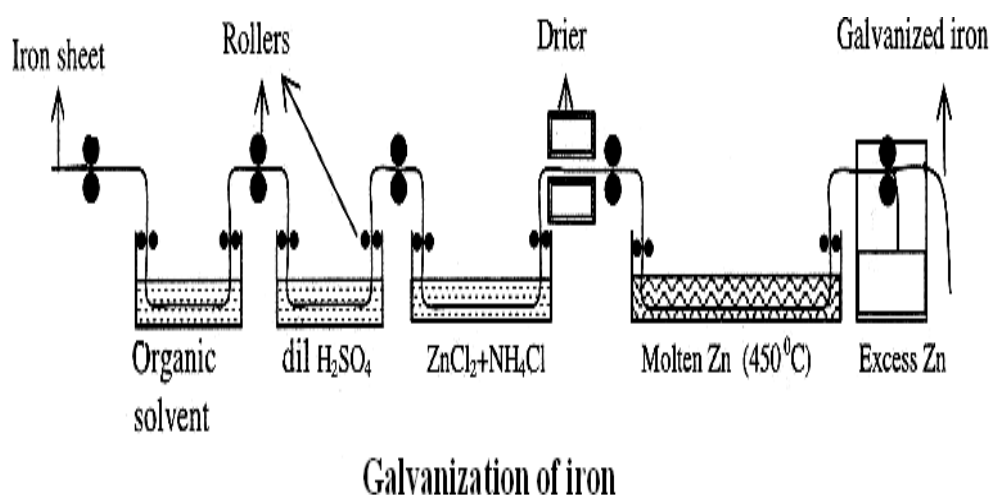
2. Cathodic protection:-1. Sacrificial Anodic Method

3. Anodizing:- Anodizing of Aluminium

Galvanizing

Coating of Zn on Fe by hot dipping process is commonly called as galvanization. It is an anodic metal coating process; which involves several steps.

- Iron sheet is passed through organic solvent to remove oil or grease present over it.
- Then, it is washed with dil. H_2SO_4 to remove any rust (Oxide layer) present on the surface.
- Then it is treated with a mixture of aqueous solution of ZnCl_2 and NH_4Cl which acts as flux and then it is dried.
- Finally it is dipped in molten zinc at 450°C .
- Excess zinc present on iron sheet is removed by passing through rollers.



Advantages:

Since the coating is anodic in nature the exposure of base metal will not cause corrosion.

Disadvantages:

Since Zn salts are poisonous food materials cannot be stored in container.

Uses:

To make galvanized articles.

In making of roofing sheets, buckets, wires, pipes.

To manufacture automobile parts, bolts & nuts, nails etc.

Anodization of Al

Defⁿ: - 'A process in which a protective oxide is formed chemically on metals by electrolysis process is known as anodization'.

Anodization of Al consists of the following steps.

Step 1: - Preparation of Al base metal surface.

The metal surface is degreased using organic solvents & rust is removed by sand blasting, grinding, acid pickling etc.

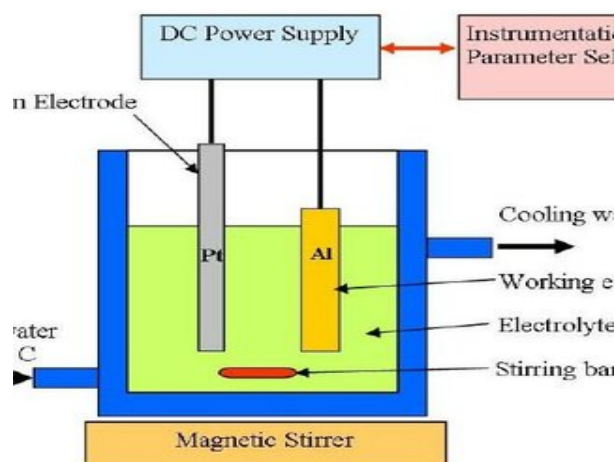
Step 2: - Al base metal is placed in an oxidizing bath like chromic acid or boric acid, H_2SO_4 .

Step 3: - The base metal is made anode and connected to a battery while lead is used as cathode. A current of $150mA/cm^2$ at 10-28V at 25-40°C is passed, Al_2O_3 gets deposited. Then it is washed with water.

Step 4: - The base metal with the oxide layer is boiled with water using colouring agent to get desired colour.



Step 5: - The base metal is further treated with nickel acetate to increase its resistance to corrosion.



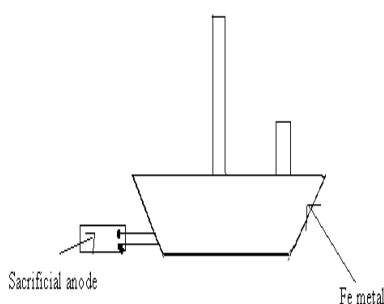
Application:

It is

It is used to make soapboxes, Tiffin carriers, window frames etc.

Cathodic protection:

The principle of this method is to convert the corrosive base metal into a cathode. And no part of it allowed to acts as anode. This is done by Sacrificial anodic protection.

Sacrificial anode method:

The metallic structure to be protected is made relatively a cathode by using another anodic metal contact with it. So that the corrosion is now concentrated on more anodic metal protecting the base metal from corrosion. The sacrificial anodes to be replaced by fresh ones as and when it is required. Commonly used sacrificial anodes are: Mg, Zn, Al etc.

This method is generally employed for the structures buried in water or streamers anchored in the sea, buried pipelines, ship hulls etc.

Corrosion Penetration Rate (CPR)

The corrosion penetration rate (CPR) is defined in three ways:

- ❖ The speed at which any metal in a specific environment deteriorates due to a chemical reaction in the metal when it is exposed to a corrosive environment.
- ❖ The amount of corrosion loss per year in thickness.
- ❖ The speed at which corrosion spreads to the inner portions of a material.

The CPR is calculated as follows:

$$\text{CPR} = (k \times W) / (D \times A \times T)$$

where k = constant

W = Total weight lost

T = Time taken for the loss of metal

A = The surface area of the exposed metal

D = The metal density in g/cm²

Numerical Problems

1. A piece of corroded steel plate was found in submerged ocean vessel. It was estimated original area of plate was 10 inch² and weight lost 2.6 kg .and assuming a CPR of 200 mils/year for this alloy in sea water. Estimate the time of submersion in year. The density of steel is 7.9 g k .

Ans: A= 10 inch² , W= 2.6 kg = 2.6X10³X10³ mg, CPR= 200 mpy t= ?

$$\text{CPR} = (k \times W) / (D \times A \times T)$$

$$200 = 534 \times 2.6 \times 10^6 / 7.9 \times 10 \times t$$

$$t = 534 \times 2.6 \times 10^6 / 7.9 \times 10 \times 200$$

$$t = 87873.41 \text{ hrs.}$$

$$\text{Therefore } t = 87873.41 / 365 \times 24$$

$$\mathbf{t = 10.3 \text{ years.}}$$

2. Calculate the CPR in both mpy and mmpy for steel sheet of area 150 inch² which is experienced a weight loss of 490 g after 1 year. Given density of steel = 7.9 g / cc

$$\text{CPR} = (k \times W) / (D \times A \times T)$$

$$= 534 \times 490 \times 1000 / 7.9 \times 150 \times 365 \times 24 \text{_____}$$

$$\mathbf{CPR = 25.2066 \text{ mpy}}$$

$$\text{CPR} = (k \times W) / (D \times A \times T)$$

$$= 87.6 \times 490 \times 1000 / 7.9 \times 150 \times 6.45 \times 365 \times 24 \text{_____}$$

$$\mathbf{CPR = 0.641 \text{ mmpy}}$$

E-waste Management

Waste can take any form that is solid, liquid [or] gas. Waste is any substance which is discarded after primary use.

Ex:- Municipal solid waste, Hazardous waste, waste water, radioactive waste, etc...

Waste management is a collection, Transportation & disposal of waste.

OR

Waste management is defined as “all the activities & Actions required to manage waste from its collections to its final disposal.

It includes

- Collection
- Transport
- Treatment
- Disposal of waste

Simple Tips to reduce waste:-

- Use of reusable bags in stores.
- Reduce the use of paper plates & cups.
- Minimize use of plastic bags & Polystyrene foam.
- By donating unwanted, slightly used clothing, furniture & other household items to local non-profit organizations.
- Purchase food in bulk [or] those which use less packaging.
- Purchase fruits that are pre-packaged in plastics.

Types of Waste Management

- ❖ Solid Waste Management
- ✓ E- Waste Management
- ❖ Bio- Medical Waste Management

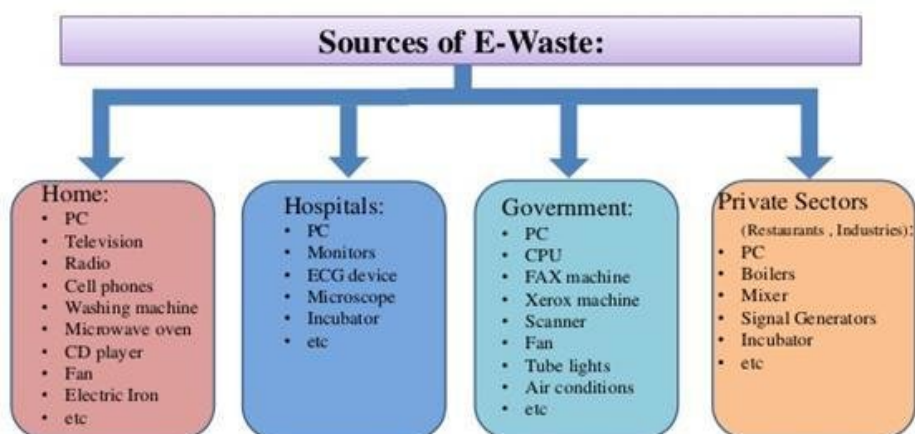
E- Waste Management

Electronic waste, popularly known as e-waste.

It is defined as electronic products that have become unwanted, on-working and have essentially reached the end of their useful life.

SOURCES and Types of E- waste

- Electronic devices such as TV's, computer monitors, laptops and display devices.
- Telecommunication devices such as cellphones, calculators, audio and video devices, printers, scanners, fax machines etc.....
- Electronic components such as sensors, alarms, sirens, security devices automobile electronic devices.
- Kitchen equipments (coffee makers, microwave ovens)



Effects of E-waste on Environment and Human Health

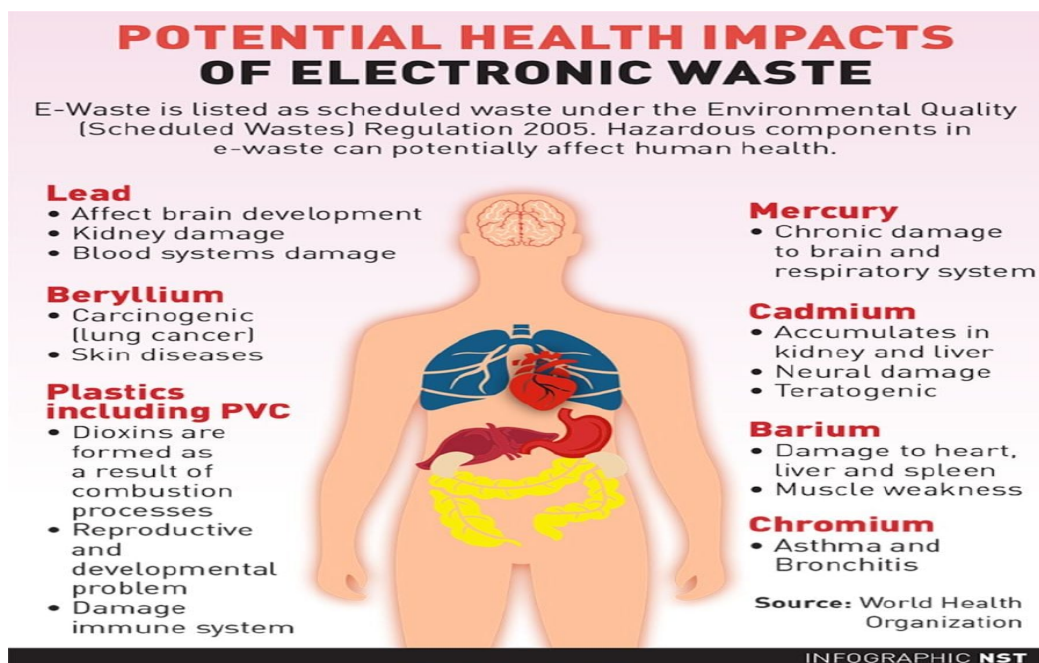
Effects of E-waste

On Environment

- Pollution of Ground-Water.
- Acidification of soil.
- Air Pollution.
- E-Waste accounts for 40 percent of the lead and 75 percent of the heavy metals found in landfills.

On Human Health

- DNA damage.
- Lung Cancer.
- Damage to heart, liver and spleen.
- Chronic damage to the brain.
- Asthmatic bronchitis.



Methods of disposal of E-Waste

The following methods are used for disposal of E-waste

- Land filling
- Incineration
- Recycling of e-waste
- Reuse

Land filling:- This is the most common methodology of e-waste disposal.

- Soil is excavated and trenches are made for burying the e-waste in it.
- It is not an environmentally sound process for disposing off the e-waste as toxic substances inside the soil and ground water.

Incineration

- This is a controlled way of disposing off the e-waste.
- It involves combustion of electronic waste at high temperature in specially designed incinerators.

Recycling of e-waste: -

- It involves the re-reutilization of e-wastes with the help of the recycling process.
- This method involves dismantling of the electronic devices, separation of the parts having hazardous substances like PCB'S and then recovery of the precious metals like copper, gold and lead can be done with the help of a e-waste recycler.

Reuse: -

- This is the most desirable e-waste recycling process where with modifications the mobile phones, computers, laptops, printers can be reused or given as second hand product.

Advantages of Recycling of E-Waste

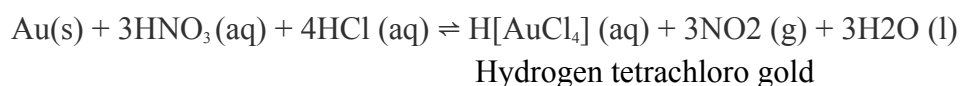
- Helps to reduce air pollution and environmental protection
- Helps to conserve available natural resources
- Creates Employment
- Increases Soil fertility and maintain Nutrients
- Helps to build a strong Economy

Extraction of Gold from E-waste (Acid Treatment)

- It is one of the easiest ways to recover gold from e-waste

Principle:

The mixture of hydrochloric acid (HCl) and nitric acid (HNO₃) have found to be useful chemicals in the extraction of gold from e-waste

**Procedure:**

- ❖ Gold was extracted using a hydrometallurgical method in six steps:
 - Grinding of CPUs
 - Leaching (waste are washed) in nitric acid
 - Leaching in aqua regia (Mixture of 3:1 HCl and HNO₃)
 - Removing the excess nitric acid
 - Extracting the gold
 - Washing and purifying the gold deposits

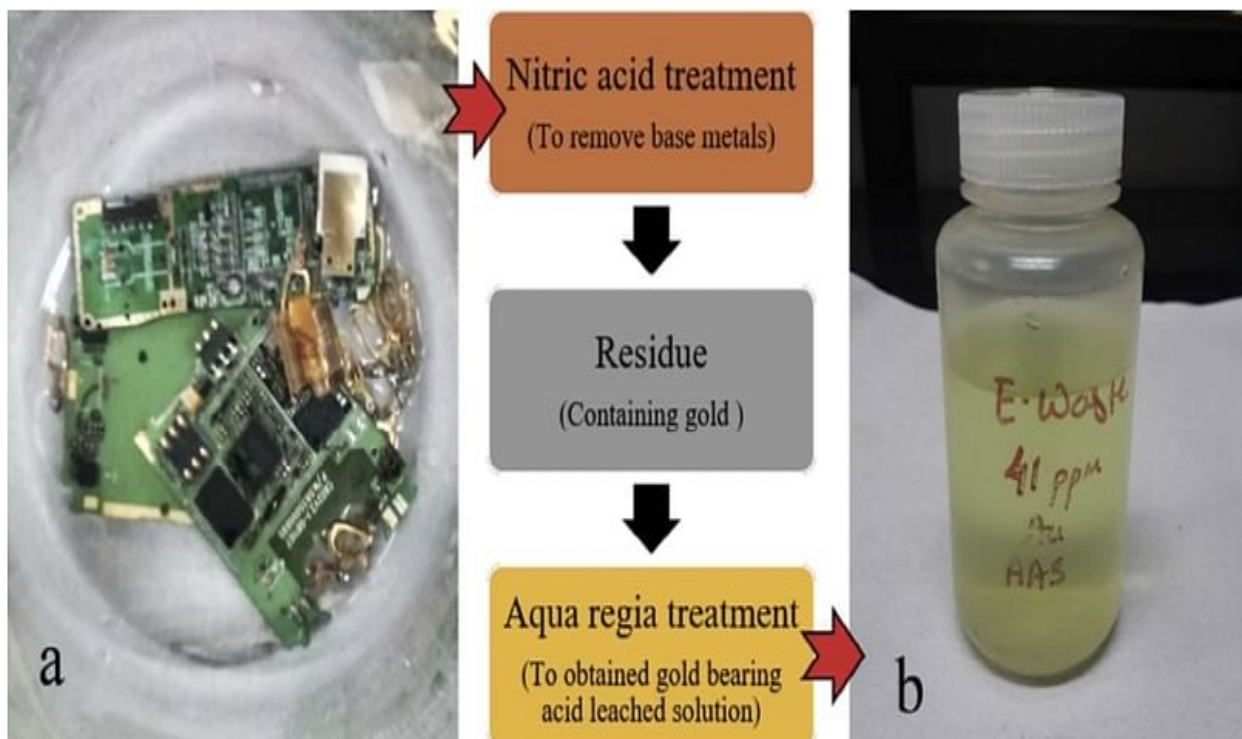
Advantages:

- High yield expected.
- Faster extraction of gold is possible.

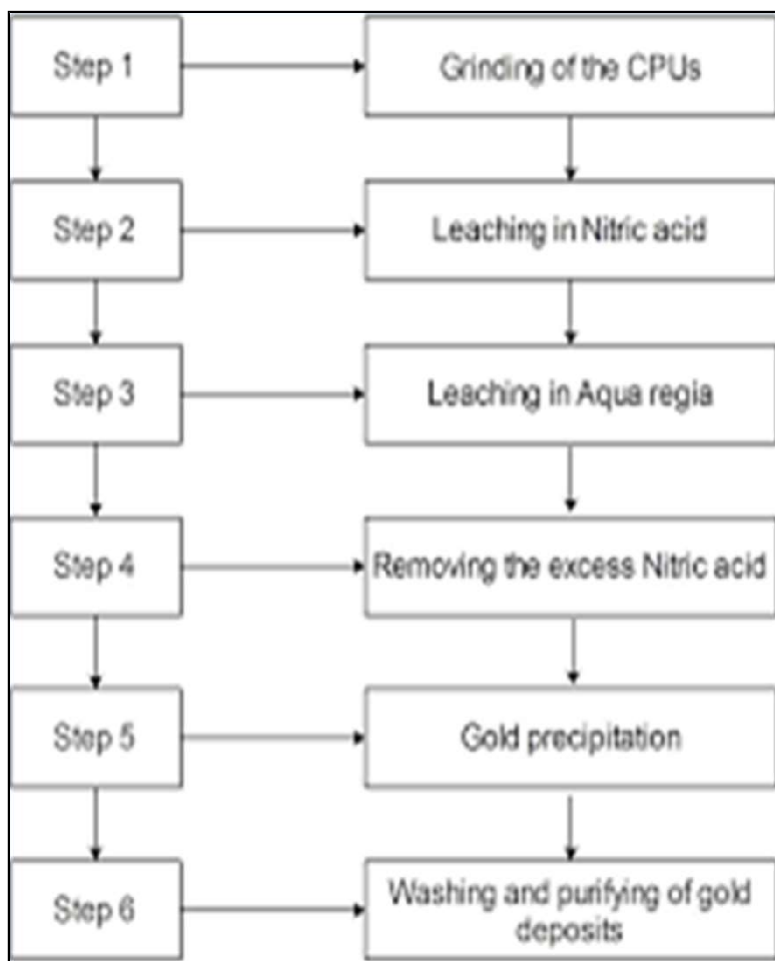
Disadvantages:

- Chemicals are used.

Acid residues are not disposed of properly, which harms the environment

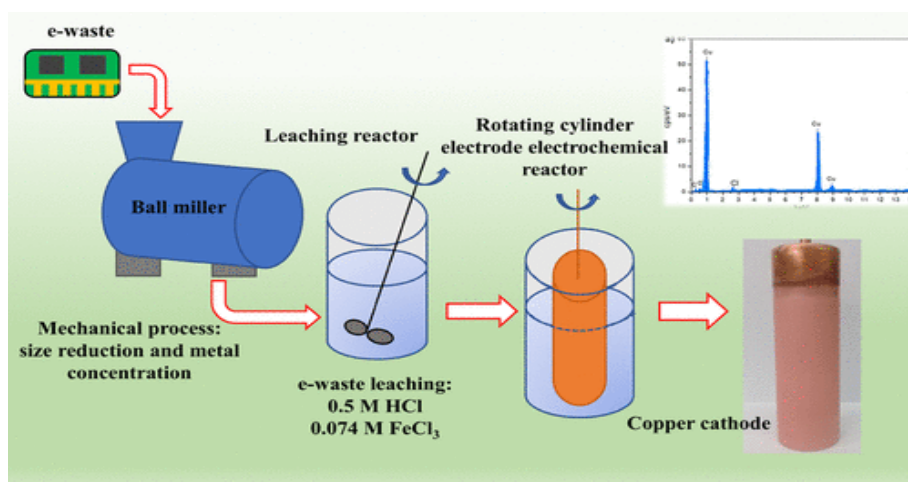


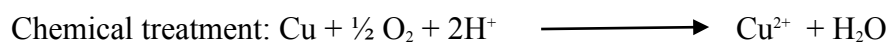
Flow Chart



Extraction of Copper from E-Waste

The hydrometallurgical process primarily involves the use of chemical reagents, where strong acids are used to leach base metals while chemical reagents such as cyanide, halide, thiourea, and thiosulfate are used to leach out precious metals from e-waste





Finally the Copper metal is recovered by electro deposition process.

MOST REPEATED QUESTIONS IN VTU EXAM

1. What is corrosion? Explain the electrochemical theory of corrosion taking iron as example.(7Marks)
2. Explain the Galvanizing process with a neat diagram.(6Marks)
3. What is Cathodic protection? Explain sacrificial anodic method. (6Marks)
4. Explain i) Differential metal corrosion ii) Waterline Corrosion ii) Pitting Corrosion. (6Marks)
5. Numerical problems on CPR .(7Marks)
6. What is E-waste? Discuss the sources, Types and effects of E-waste on Environment and Human Health. (7Marks)
7. Explain the Methods of E-waste disposal .(7Marks)
8. Explain the Extraction of Gold from E-waste. (6Marks)
9. Explain the Extraction of Copper from E-waste. Mention the any 2 advantages of recycling. .(7Marks)
